

ASHISH KUMAR, PhD

Materials Scientist & Research Faculty

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PROFESSIONAL SUMMARY

Materials Scientist and Postdoctoral Researcher with 8+ years of expertise in advanced functional polymer fiber/membrane engineering, energy storage safety, separation science, and wet-chemistry processing (electrospinning, solution blowing, slurry ink formulation). Proven track record leading US EPA-, NIOSH-, and industry-sponsored grants across battery thermal runaway diagnostics (NMC/LFP/NCA), heavy metal/acid mine drainage remediation, and surfactant-electrolyte transport dynamics. Highly proficient in advanced analytical metrology (SEM/TEM, XRD, XPS, KPFM, TGA/DSC, EIS), data-driven materials informatics (Python/MATLAB, ANNs), federal grant writing, and cross-functional team leadership across academia and Fortune 500 corporate partners (**Freepport-McMoRan**, **BASF**, **Clariant**, **Innospec**, **AFI India**, **Elofic**).

TECHNICAL SKILLS & CORE COMPETENCIES

- **Microstructural & Surface Metrology:** Transmission Electron Microscopy (TEM, HR-TEM), Field Emission Scanning Electron Microscopy (FESEM/EDS), X-ray Diffraction (XRD, XRPD), Atomic Force Microscopy (AFM), Kelvin Probe Force Microscopy (KPFM Surface Potential), X-ray Photoelectron Spectroscopy (XPS), In-Situ FTIR, Raman, UV-Vis, Dynamic Light Scattering (DLS), Zeta Potential, Contact Angle Goniometry.
- **Advanced Manufacturing & Processing:** Scalable Nanofiber Fabrication (Electrospinning, Supersonic Solution Blowing, Free-Surface Unconfined Melt Electrospinning), Polymer Processing, Surfactant-Electrolyte Chemistry, Micro/Nano Heterostructures, Slurry Formulation, 2D Nanomaterial Exfoliation ($\text{Ti}_3\text{C}_2\text{T}_x$ MXene, rGO).
- **Thermal, Mechanical & Battery Testing:** Universal Testing Machine (UTM – Tensile/Stress-Strain), Differential Scanning Calorimetry (DSC), Thermogravimetric Analysis (TGA/DTG), High-Temperature Thermal Abuse Diagnostics, Battery Off-Gas Dynamic FTIR Tracking, High-Speed Optical Imaging (Phantom v2512).
- **AI/ML, Data Science & Modeling:** Materials Discovery Informatics (Python, TensorFlow, Artificial Neural Networks – ANNs, Gaussian Process Regression, Bayesian Optimization), MATLAB, R, MySQL, Multiphase Transport Modeling.
- **Project Leadership & Grant Writing:** Drafting NSF Proposals (Materials Discovery, Waste Valorization, Surfactant Adsorption, Acoustic Mitigation), Ministry of Textiles Research Grants, SOP Formulation, Quality Management Systems (ISO 9001 Awareness), Electronic Laboratory Notebook (ELN) Documentation.

EDUCATION

- **Indian Institute of Technology Mandi** **Mandi, India**
2017 – 2022
Ph.D. in Mechanical and Materials Engineering (CGPA: 8.25/10)
 - *Dissertation:* Environmental Pollution Mitigation using Solution Blown Nanofibers.
 - Invented supersonic aerodynamic blowing technology converting plastic waste into functional filtration media (**Granted Indian Patent No. 482026**).
- **National Institute of Technology Kurukshetra** **Kurukshetra, India**
2015 – 2017
M.Tech. in Material Science and Nanotechnology (CGPA: 8.5/10)
 - *Thesis:* Fabrication & Characterization of Heterostructures & High Electron Mobility Transistors toward Sensing Applications (in collaboration with CSIR-CSIO, Chandigarh).
- **Lingaya's University** **Faridabad, India**
2011 – 2015
B.Tech. in Electronics and Communication Engineering (CGPA: 7.02/10)
 - *Capstone Project:* Design and evaluation of silicon-based solar power conversion device.

RESEARCH EXPERIENCE

- **Chitkara University, CURIN Department** **Punjab, India**
2026 – Present
Research Faculty
 - Leading cutting-edge research at the Centre for Innovations to foster full-cycle processing of agriculture waste to wealth (PAWW).
- **Missouri University of Science and Technology** **Rolla, USA**
2023 – 2026
Postdoctoral Research Fellow – Advanced Materials & Energy Technologies
 - Led research operations on **US EPA Grant No. 97797001** and corporate R&D projects sponsored by global industrial leaders (**Freepport-McMoRan**, **BASF**, **Clariant**, **Innospec**).
 - Investigated surfactant-electrolyte interfacial transport, bubble coalescence, and aerosol ejection dynamics to suppress sulfuric acid mist in copper electrowinning (*J. Sustainable Metallurgy*, 2025; *J. Electrochem. Soc.*, 2026).
 - Evaluated structural degradation, flame dynamics, and toxic off-gas release (HF, CO) in energy storage battery modules under **NIOSH Grant No. 1U60OH012350-01-00** (*J. Power Sources*, 2025).
 - Engineered electroactive PVDF/ $\text{Ti}_3\text{C}_2\text{T}_x$ MXene and rGO functional nanofibrous matrices, utilizing KPFM surface potential mapping for high-efficiency submicron aerosol capture (*Environ. Pollut.*, 2026; *ACS Omega*, 2026).
 - Deployed AI/ML data-driven algorithms in Python/MATLAB for property prediction, active learning, and automated spectral deconvolution of complex functional nanomaterials.

- Maintained analytical characterization facilities (FESEM, XRD, FTIR, UTM), drafted SOPs, and co-mentored 7 Ph.D. students.
- **Indian Institute of Technology Delhi** **Delhi, India**
2022 – 2023
Senior Research Fellow – Functional Nanomaterials & Filtration Technology
 - Partnered directly with commercial industrial manufacturers (**AFI India, Elofic Industries**) under DSIR-funded projects to optimize fluid-particle transport and structural durability of porous media architectures.
 - Developed Artificial Neural Network (ANN) and fluid transport models in Python to predict system separation efficiency, energy loads, and indoor environmental metrics.
 - Authored technical compliance reports, standardized testing protocols, and technology transfer documentation for pilot scale-up.

SELECTED FUNDED R&D PROJECTS

- **Extractive Metallurgy & Environmental Safety Programs** **2023 – 2026**
Missouri S&T | US EPA, NIOSH & Corporate Consortia
 - **EPA Grant No. 97797001:** Quantitative evaluation of surfactant-electrolyte interactions and interfacial bubble dynamics for acid mist suppression in copper electrowinning (**Freeport, BASF, Clariant**).
 - **NIOSH Grant No. 1U60OH012350-01-00:** Fire-induced aerosol/gas generation, ventilation interactions, and emergency response in battery electric vehicle incidents.
 - **Corporate R&D (InterCom Global, USA):** Characterization of airborne emissions, thermal abuse, and safety risks from lithium-ion battery thermal runaway events.
 - **Mining Aerosol Mitigation (MST-SEED Grant):** Monitoring, characterization, and filtration of respirable coal dust and crystalline silica dust using electrospun nanofibers.
- **Smart Separation, Sustainable Water Treatment & Clean Energy** **2017 – 2023**
IIT Mandi & IIT Delhi | MHRD, SERB, DSIR & Industry Programs
 - **DSIR Industry Collaborative Project (IIT Delhi):** Post-COVID-19 central air-conditioning and ventilation optimization guided by AI/ML algorithms (**AFI India, Elofic Industries**).
 - **Project ID: 15/18-PERC/2018-19 (IIT Mandi):** Treatment of Acid Mine Drainage via Chitosan/Nylon-6 biopolymer nanocomposites for heavy metal removal (*Polymer*, 2021).
 - **SERB Project (IITM/SERB/SSR/215):** High-efficiency capture of most penetrating particles (350 nm) using supersonically blown ultrafine PVDF nanofibers.
 - **MHRD IMPRINT (IITM/MHRD-IMPRINT/AD/169):** Sustainable wastewater treatment through bio-photoelectric catalysis and bioproduction.
 - **IIT Mandi Seed Grants:** Development of electrochemical biosensing platforms for environmental analytes and indigenous protective media from recycled polymer waste.
- **Solid-State Heterostructures & Photovoltaics** **2011 – 2017**
NIT Kurukshetra & Lingaya's Univ. | CSIR-CSIO Chandigarh & Capstone
 - Fabrication and characterization of semiconductor heterostructures and High Electron Mobility Transistors (HEMTs) for electrical sensing applications.
 - Synthesis of Ag-CdSe nanocomposites for optoelectronic systems; design and prototyping of a silicon-based solar charger.

TEACHING, PEDAGOGICAL EXPERIENCE & MENTORSHIP

- **Missouri University of Science and Technology** **Rolla, USA**
2023 – 2026
Postdoctoral Research Mentor
 - Mentored 7 Ph.D. students in experimental setup design, high-speed imaging analysis, aerosol metrology, and manuscript drafting.
- **Indian Institute of Technology Delhi** **Delhi, India**
2022 – 2023
Research Mentor
 - Mentored 1 Ph.D., 1 Master's, and 2 Undergraduate students in developing ANN models for HVAC performance and filtration analytics.
- **Indian Institute of Technology Mandi** **Mandi, India**
2017 – 2022
Teaching Assistant & Mentor
 - Served as Teaching Assistant for undergraduate and graduate courses: *Material Science for Engineers* (IC-241), *Nanomanufacturing* (ME-509), *Mechanical Workshop*, *Creep-Fatigue Interactions* (EN-613), and *Manufacturing Engineering* (ME-308).
 - Prepared curricula, conducted lab sessions, designed mid-term/end-term exams, and provided individual tutorials.
 - Received the **Best Teaching Assistant Award** for the *Material Science for Engineers* course.
 - Mentored 3 Ph.D. candidates, 3 Master's students, and 1 Undergraduate student in experimental materials research.
- **National Institute of Technology Kurukshetra** **Kurukshetra, India**
2015 – 2017
Graduate Teaching Assistant
 - Conducted laboratory practicals and tutorial sessions for *Nanomaterials and their Properties* (MNT501T), *Material Characterization Techniques* (MNT507T), and *Nano Electronics* (MNT506T).

PATENTS & INTELLECTUAL PROPERTY

- **Granted Patent (27/05/2023):** Nanofiber from waste Plastic bottles. Patent No: **482026** (App. No: 20201102217). 
- **Granted Patent (08/09/2026):** Nanofiber Air filter derived from waste cigarette butts and a method for preparing the same. Patent No:

PEER-REVIEWED JOURNAL PUBLICATIONS

- [J1] **Kakoria, A.**, Zaid, M.M., Iqbal, A., Afful, E.A. and Xu, G., 2025. Optimizing Acid Mist Suppression: Unraveling Surfactant Effects on Bubble Formation and Bursting Dynamics in Copper Electrowinning. *Journal of Sustainable Metallurgy*, pp.1–12. [🌐]
- [J2] **Kakoria, A.**, Zaid, M.M. and Xu, G., 2026. Sulfuric Acid Mist Generation, Calculation and Suppression Mechanism and Methods in Electrowinning: A Review. *Journal of The Electrochemical Society*. [🌐]
- [J3] **Kakoria, A.** and Sinha-Ray, S., 2018. A review on biopolymer-based fibers via electrospinning and solution blowing and their applications. *Fibers*, 6(3), p.45. [🌐]
- [J4] **Kakoria, A.**, Chandel, S.S. and Sinha-Ray, S., 2021. Novel supersonically solution blown nanofibers from waste PET bottles for PM0.1-2 filtration: From waste to pollution mitigation. *Polymer*, 124260. [🌐]
- [J5] **Kakoria, A.** and Sinha-Ray, S., 2022. Ultrafine nanofiber-based high-efficiency air filter from waste cigarette butts. *Polymer*, 125121. [🌐]
- [J6] **Kakoria, A.**, Zaid, M.M., Iqbal, A., Moradi, M., Zheng, L. and Xu, G., 2026. Electroactive PVDF/Ti₃C₂T_x MXene Functionalized Nanofiber Mats with High Surface Potential for High-Efficiency Indoor & Outdoor Submicron PM (0.3–2) μ m Filtration. *Environmental Pollution*, p.127659. [🌐]
- [J7] **Kakoria, A.**, Devi, B., Anand, A., Halder, A., Koner, R.R. and Sinha-Ray, S., 2018. Gallium oxide nanofibers for hydrogen evolution and oxygen reduction. *ACS Applied Nano Materials*, 2(1), pp.64–74. [🌐]
- [J8] **Kakoria, A.**, Sinha-Ray, S. and Sinha-Ray, S., 2021. Industrially scalable Chitosan/Nylon-6 (CS/N) nanofiber-based reusable adsorbent for efficient removal of heavy metal from water. *Polymer*, 213, p.123333. [🌐]
- [J9] Sheoran, N., Shabani, E., Priya, R., Adhikari, J. and **Kakoria, A.**, 2026. Free surface unconfined melt electrospinning: an emergent approach. *Journal of Materials Science*, pp.1–23. [🌐]
- [J10] Zaid, M.M., Kumar, A.R., **Kakoria, A.**, Zheng, L. and Xu, G., 2026. Advancements in fibrous aerosol filtration across experimental analysis, numerical modeling, and fabrication technologies. *Journal of Materials Science*, pp.1–36. [🌐]
- [J11] Moradi, M., **Kakoria, A.**, Zheng, L. and Xu, G., 2026. Efficiency Comparison of Reduced Graphene Oxide (rGO) Composite-Incorporated Polyacrylonitrile Nanofibrous Filters for Respirable Crystalline Silica Dust and Coal Dust. *ACS Omega*. [🌐]
- [J12] Thakur, S.S., Chandel, S.S., **Kakoria, A.** and Sinha-Ray, S., 2022. Enhancement in pool boiling heat transfer of ethanol and nanofluid on novel supersonic nanoblown nanofiber textured surface. *Experimental Heat Transfer*, 35(4), pp.516–532. [🌐]
- [J13] Iqbal, A., **Kakoria, A.**, Riaz, S.T., Xu, J., Pushparaj, R.I. and Xu, G., 2025. Comparative fire hazards of lithium-ion battery chemistries: Linking thermal behavior, gas toxicity, and state-of-charge to composite risk profiles. *Journal of Power Sources*, 655, p.237914. [🌐]
- [J14] Zaid, M.M., Amoah, N., **Kakoria, A.**, Wang, Y. and Xu, G., 2023. Advancing occupational health in mining: investigating low-cost sensors suitability for improved coal dust exposure monitoring. *Measurement Science and Technology*, 35(2), p.025128. [🌐]
- [J15] Xu, J., Tian, R., **Kakoria, A.**, Iqbal, A., Yang, T., Zheng, L. and Xu, G., 2026. Underground mine tunnel fire dynamics: Experimental investigation and AI-driven characterization of flame behavior. *Tunnelling and Underground Space Technology*, 174, p.107719. [🌐]

PEER-REVIEWED CONFERENCE PROCEEDINGS & BOOK CHAPTERS

- [C1] **Kakoria, A.**, Zaid, M.M., Iqbal, A. and Xu, G., 2026, June. PVDF–Ti₃C₂T_x (MXene) Ultrafine Nanofiber Membrane for Effective Fine Particulate Matter (0.1–2.5 μ m) Filtration. In *North American Mine Ventilation Symposium 2025* (p. 405). CRC Press. [🌐]
- [C2] Riaz, S.T., Iqbal, A., **Kakoria, A.** and Xu, G., 2026, June. Underground Mines Using the Analytic Hierarchy Process. In *North American Mine Ventilation Symposium 2025* (p. 82). CRC Press. [🌐]
- [C3] Giri, P., **Kakoria, A.**, Verma, S. and Sinha-Ray, S., 2022. Nanofibers for sustainable filtration: a waste to energy approach. In *Machines, Mechanism and Robotics: Proceedings of iNaCoMM 2019* (pp. 1693–1701). Springer Singapore. [🌐]
- [C4] Giri, P., **Kakoria, A.**, Verma, S. and Sinha-Ray, S., 2020. Reuse of Cigarette Filters for Energy Applications. In *Functional Textiles and Clothing 2020* (pp. 161–168). Springer Singapore. [🌐]

MANUSCRIPTS UNDER REVIEW & IN PREPARATION

- **[Under Review]** Laboratory Investigation of Various Extinguishing Agents on Cylindrical Lithium-Ion Batteries with NMC and NCA Cathode Chemistries – *Journal of Energy Storage*.
- **[Under Review]** Linking Fabrication Route and KPFM Surface Potential Distribution to Submicron Filtration Performance of Additive-Free Recycled PET Nanofibers – *Journal of Separation and Purification Technology*.
- **[Under Review]** Characterizing Surface Potential Distributions of Electrospun and Supersonic Solution-Blown Recycled PET Nanofiber Membranes by Full-Area KPFM – *Polymer*.
- **[Under Review]** MXene Loading Dependent Surface Potential Distribution and Electrostatic Particle Capture in Electrospun Nanofiber Membranes – *Polymer*.
- **[Under Review]** Effects of State of Charge and Scale on Thermal Runaway, Gas Emissions, and Fire Risk in Lithium-Ion Battery Modules – *Journal of Energy Storage*.
- **[In Prep.]** Quantitative Analysis of Gases Emitted During LIB Thermal Runaway: A comparative study of SOC effects on LFP and NMC modules – *Journal of Power Sources*.
- **[In Prep.]** Optimizing the Impacts of Organic and Surfactants on the Suppression of Sulfuric Acid Mist in Electrowinning Systems – *Hydrometallurgy*.
- **[In Prep.]** Advancing Aerosol Filtration: Theoretical Foundations, Computational Models, and Experimental Insights – *Separation*

and Purification Technology.

- **[In Prep.]** Synergistic Charge Engineered Supersonically Nanoblown Ultrafine Nanofibers for the Advanced Filtration of Respirable Silica Dust – *Environmental Science & Technology*.

INVITED TALKS & INTERNATIONAL PRESENTATIONS

- **Ashish Kakoria**, Mirza Mohammad Zaid, and Guang Xu. “A Study on the Effect of Surfactant on the Bubble Size, its Bursting Dynamics, and Acid Mist Suppression Using High-Speed Camera Imaging.” *SME Annual Conference*, Feb 23–26, 2025, Denver, CO, **USA**.
- **Ashish Kakoria**, Ellen Amoako Afful, and Guang Xu. “Acid mist pollution prevention for copper electrowinning facilities.” *SME Arizona Conference*, Dec 8, 2024, Tucson, AZ, **USA**.
- **Ashish Kakoria**, and Guang Xu. “Controlling acid mist for copper electrowinning.” *7th International Symposium on Mine Safety Science and Engineering*, Aug 18, 2024, Pittsburgh, PA, **USA**.
- **Ashish Kakoria**, Michael Moats, and Guang Xu. “Controlling acid mist for copper electrowinning.” *SME Arizona Conference*, Dec 3, 2023, Tucson, AZ, **USA**.
- **Ashish Kakoria**, Mirza Mohammad Zaid, and Guang Xu. “Effect of Surfactants on the bubble size, and its bursting mechanism in copper electrowinning using high speed camera imaging.” *SME Annual Conference*, Feb 25–28, 2024, Phoenix, AZ, **USA**.
- **Ashish Kakoria**, Mirza Mohammad Zaid, and Guang Xu. “PVDF–Ti₃C₂T_x (MXene) ultrafine nanofiber membrane for effective fine particulate matter filtration.” *International Mine Ventilation Conference (IMVC)*, Aug 11–15, 2024, Sydney, **Australia**.
- **Ashish Kakoria**, and Guang Xu. “Nanofiber Filters for Industrial Dust Control.” *20th North American Mine Ventilation Symposium*, Jun 21–26, 2024, Pittsburgh, PA, **USA**.
- **Ashish Kakoria**, and Sumit Sinha Ray. “Novel supersonically solution blown nanofibers from waste PET bottle for PM0.1-2 filtration.” *International Workshop on Advanced Materials (IWAM)*, Feb 22–26, 2022, **Dubai**, UAE.
- **Ashish Kakoria**, and Sumit Sinha Ray. “Industrially scalable Chitosan/Nylon-6 (CS/N) nanofiber-based reusable adsorbent for efficient removal of heavy metal from water.” *International Workshop on Advanced Materials (IWAM)*, Feb 22–26, 2020, **Dubai**, UAE.
- *National Presentations (India)*: FTC-2020 (IIT Delhi), HIMCOSTE-2020 (IIT Mandi), ICT-2020 (ICT Mumbai), ICFNM-2019 (IIT BHU), NCRDNN-2019 (Jadavpur Univ.), ICANN-2019 (IIT Guwahati).

RESEARCH GRANT WRITING & PROPOSALS

- Co-authored NSF Proposal: *Molecular Structure Directed Surfactant Adsorption at the Electrolyte-Air Interface: Implications for Enhanced Acid Mist Suppression in Copper Electrowinning*.
- Co-authored NSF Proposal: *Using Low-Frequency Sound Waves to Suppress Acid Mist in Copper Electrowinning*.
- Co-authored NSF Proposal: *Valorization of PET waste into ultrafine nanofiber-based gas sensor and filter to mitigate toxic gas emission from lithium-ion batteries*.
- Co-authored NSF Proposal: *Using AI and Alternative Approaches for Nanofiber Material Discovery*.
- Prepared Ministry of Textiles Proposal: *Multi-metal and hetero atom decorated energy efficient carbon nanofibers from spent heavy metal membranes*.

HONORS, AWARDS & PROFESSIONAL SERVICE

- Doctoral research on plastic waste upcycling featured across **200+ media channels** (Times of India, NDTV, Mint, Dainik Bhaskar).
- **Best Teaching Assistant Award** for *Material Science for Engineers* course (IIT Mandi).
- International Travel and Accommodation Grants – IWAM Dubai (2020, 2022).
- **Best Prototype Award** on Engineer’s Day; Qualified National GATE Examination.
- **Peer Reviewer**: Elsevier Journals (*Process Biochemistry*, *International Journal of Coal Science & Technology*).

REFERENCES

- **Dr. Guang Xu** | *Postdoctoral Advisor*
Associate Professor, Department of Mining and Explosives Engineering, Missouri S&T, USA
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- **Prof. Amit Rawal** | *Postdoctoral Advisor*
Professor, Department of Textile and Fibre Engineering, Indian Institute of Technology Delhi, India
Email: Amitrawal77@hotmail.com | Phone: +91-9958205917
- **Dr. Ashish R. Kumar** | *Research Collaborator / Mentor*
Assistant Professor, Department of Energy and Mineral Engineering, Penn State University, USA
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